

Coding & Solution

Date

06 July 2026

Team ID

Project Name

Smart Lender — Applicant Credibility Prediction for Loan Approval

Maximum Marks

5 Marks

Solution Summary

Field	Details
Repository Link / URL	
Programming Language(s)	Python (backend, ML), HTML/CSS/JavaScript (frontend)
Framework(s) Used	Flask, scikit-learn, XGBoost, imbalanced-learn
Key Features Implemented	Loan eligibility prediction form; real-time prediction via trained XGBoost model; 4-model comparison pipeline; SMOTE-based class balancing; styled, animated result page
Pending / Incomplete Features	Cloud deployment (currently runs on local Flask dev server); persistent database for applicant history
Setup / Run Instructions	1) pip install -r requirements.txt 2) Ensure loan_prediction.csv, rdf_model.json, scale1.pkl are present 3) python app.py 4) Open http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes (once pushed to GitHub)

Additional Notes / Comments

The model is saved using XGBoost's native JSON format rather than pickle, after an earlier pickle-based save caused an 'input stream corrupted' error when the model was loaded on a different machine/XGBoost version. This change made the deployment portable across environments.